Unit 6 Review

Relationships in Triangles

Name_____
Date_____
Period_____

Show your work. Circle your final answer.

HOW TO USE THIS. This review covers every learning target of Lessons 1–8 — more than the test does. Work it in parts. If you miss two or more in a part, that part is where to spend your study time. The reference sheet has everything you need for each part. Thread B's two angle days come after the test and are not on it.

PART A. The Triangle Inequality and the Converse

Lessons 1–2 · 4 questions

1. Can the three lengths be the sides of a triangle? Show the test.

a. 5 cm, 9 cm and 15 cm

b. 6 inches, 10 inches and 16 inches

c. 11 m, 4 m and 8 m

2. Two sides of a triangle are 6 ft and 14 ft. Write an inequality for every possible length x of the third side.

3. Is a triangle with these sides a right triangle? Show the test.

a. 65, 33 and 56

b. 7, 8 and 11

4. Select ALL of the sets of side lengths that make a right triangle.

- ☐ A. 6, 7, 8
- ☐ B. 10, 24, 26
- ☐ C. 3, 4, 7
- ☐ D. 8, 9, 12
- ☐ E. 16, 30, 34

PART B. The Pythagorean Theorem

Lesson 3 · 6 questions

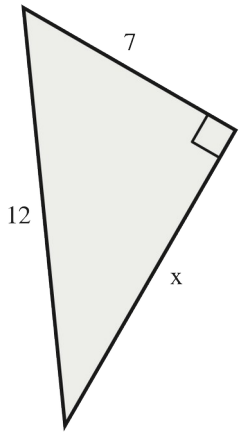
5. A right triangle has legs of 12 ft and 5 ft. How long is the hypotenuse?

6. A right triangle has legs of 3 inches and 13 inches. How long is the hypotenuse, exactly and to the nearest tenth?

7. A right triangle has a hypotenuse of 85 cm and one leg of 84 cm. How long is the other leg?

8. A right triangle has a hypotenuse of 11 ft and one leg of 8 ft. How long is the other leg, exactly and to the nearest tenth?

9. Which equation can be used to find x?



- ○ A. $7^2 + 12^2 = x^2$
- ○ B. $7 + x = 12$
- ○ C. $7^2 + x^2 = 12^2$
- ○ D. $7^2 + x^2 = 2 \cdot 12$

10. Ana finds the missing leg of a right triangle whose hypotenuse is 5 cm and whose other leg is 3 cm. She writes $3^2 + 5^2 = 34$, so the leg is about 5.8 cm. What did she do wrong? Find the leg.

PART C. The Pythagorean Theorem in Real-World Problems

Lesson 4 · 5 questions

11. A 10-ft ladder leans against a wall with its foot 6 ft from the wall. How high up the wall does it reach?

12. A rectangular garden is 10 m long and 11 m wide. How long is a path straight across it from corner to corner, to the nearest tenth of a meter?

13. A ramp rises 1 ft over a run of 9 ft along the ground. How long is the ramp, to the nearest tenth of a foot?

14. A park is a rectangle 9 m by 11 m. How much shorter is it to walk straight across the park from corner to corner than to walk along two of its sides? Give the answer to the nearest tenth.

15. A 9-ft ladder stands with its foot 2 ft from a wall. Will it reach a window ledge 8.5 ft up the wall?

PART D. Distance on the Coordinate Plane

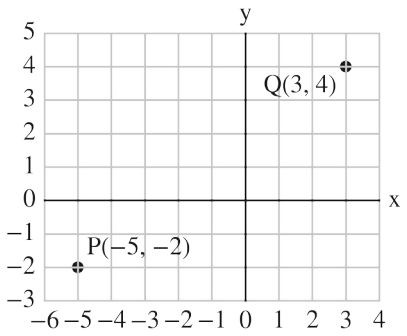
Lesson 5 · 5 questions

16. Find the distance between the two points.

a. $(-6, 2)$ and $(5, 2)$

b. $(3, -7)$ and $(3, 2)$

17. Find the distance between points P and Q.



18. Find the distance between (−6, −1) and (1, 5), to the nearest tenth.

19. How far is the point (−1, 6) from the origin? Round your answer to the nearest tenth.

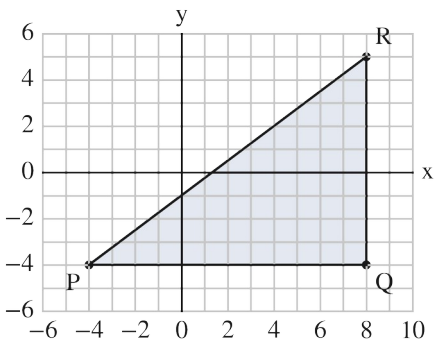
20. What is the distance between (−2, −6) and (−2, 3)?

- ○ A. 3 units
- ○ B. 9 units
- ○ C. 10 units
- ○ D. −9 units

PART E. Coordinate-Plane Problems and Perimeter

Lesson 6 · 4 questions

21. Every part below is about triangle PQR on the grid, with corners P(−4, −4), Q(8, −4) and R(8, 5).



a. Find the length of PQ.

b. Is triangle PQR a right triangle? How do you know?

c. Find the length of PR.

d. A triangle similar to PQR has a shortest side of 3. How long is its longest side?

22. Find the perimeter of the triangle with corners (−4, 2), (6, 2) and (6, −1), to the nearest tenth.

23. On a town map in blocks, the school is at (−2, −4) and the park is at (2, 1).

a. How far is it from the school to the park along the streets?

b. How far is it in a straight line? Round your answer to the nearest tenth.

24. A rectangle has corners (−3, −2), (5, −2), (5, 4) and (−3, 4).

a. Find its perimeter.

b. Find the length of one of its diagonals.

PART F. Similar Triangles and Indirect Measurement

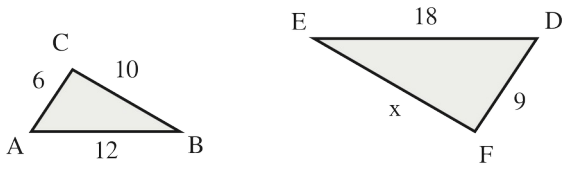
Lessons 7–8 · 6 questions

25. Are the two triangles similar? Show the three ratios.

a. Sides 5, 7, 8 and sides 15, 21, 24

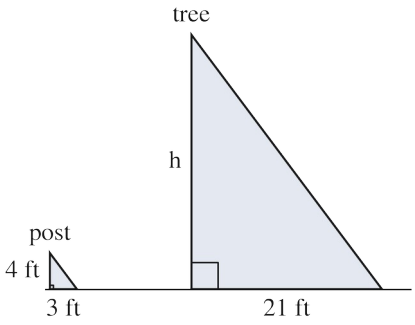
b. Sides 5, 8, 10 and sides 10, 16, 18

26. Triangle ABC is similar to triangle DEF. Find x.



27. A triangle has sides 15, 20 and 30. A similar triangle's middle side is 8. Find its other two sides.

28. A 4-ft fence post casts a 3-ft shadow. At the same moment, a tree casts a 21-ft shadow. How tall is the tree?



29. Kai's eyes are 5 ft above the ground. He stands 3 ft from a mirror on the ground and sees in it the top of a pole that stands 12 ft beyond the mirror. The two right triangles are similar. How tall is the pole?

30. Which triangle is similar to one with sides 6, 8 and 9?

- ○ A. 8, 10, 11
- ○ B. 12, 16, 20
- ○ C. 9, 12, 13.5
- ○ D. 6, 8, 12

GRADE 7 ACCELERATED · UNIT 6

Unit 6 Review

Relationships in Triangles **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

HOW TO USE THIS. This review covers every learning target of Lessons 1–8 — more than the test does. Work it in parts. If you miss two or more in a part, that part is where to spend your study time. The reference sheet has everything you need for each part. Thread B's two angle days come after the test and are not on it.

PART A. The Triangle Inequality and the Converse

Lessons 1–2 · MA.8.GR.1.3 · 4 questions

1. Can the three lengths be the sides of a triangle? Show the test.

a. 5 cm, 9 cm and 15 cm

No — 5 + 9 = 14 is less than 15.

the two shorter sides against the longest

b. 6 inches, 10 inches and 16 inches

No — 6 + 10 = 16 is equal to the longest side, so the sides lie flat.

equal is not more

c. 11 m, 4 m and 8 m

Yes — 4 + 8 = 12 is more than 11.

the longest is listed first; 11 + 4 tests the wrong pair

2. Two sides of a triangle are 6 ft and 14 ft. Write an inequality for every possible length x of the third side.

$8 < x < 20$

between the difference, 14 − 6, and the sum, 14 + 6 — never equal to either

3. Is a triangle with these sides a right triangle? Show the test.

a. 65, 33 and 56

Yes — $33^2 + 56^2 = 1{,}089 + 3{,}136 = 4{,}225 = 65^2$.

the longest, 65, is c, though it is listed first

b. 7, 8 and 11

No — $49 + 64 = 113$, and $11^2 = 121$.

a triangle (7 + 8 > 11), but not a right one

4. Select ALL of the sets of side lengths that make a right triangle.

- ☐ A. 6, 7, 8
- ☒ **B. 10, 24, 26**
- ☐ C. 3, 4, 7
- ☐ D. 8, 9, 12
- ☒ **E. 16, 30, 34**

B and E

$100 + 576 = 676 = 26^2$ and $256 + 900 = 1{,}156 = 34^2$: both, and nothing else.

PART B. The Pythagorean Theorem

Lesson 3 · MA.8.GR.1.1 · 6 questions

5. A right triangle has legs of 12 ft and 5 ft. How long is the hypotenuse?

13 ft

144 + 25 = 169

6. A right triangle has legs of 3 inches and 13 inches. How long is the hypotenuse, exactly and to the nearest tenth?

√178 in ≈ 13.3 in

9 + 169 = 178

7. A right triangle has a hypotenuse of 85 cm and one leg of 84 cm. How long is the other leg?

13 cm

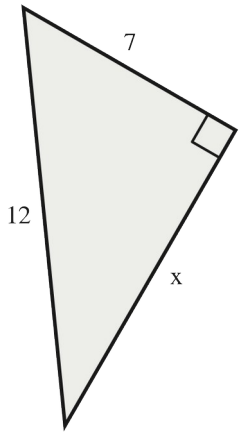
7,225 − 7,056 = 169

8. A right triangle has a hypotenuse of 11 ft and one leg of 8 ft. How long is the other leg, exactly and to the nearest tenth?

√57 ft ≈ 7.5 ft

121 − 64 = 57

9. Which equation can be used to find x?



- ○ A. $7^2 + 12^2 = x^2$
- ○ B. $7 + x = 12$
- ● **C.** $7^2 + x^2 = 12^2$
- ○ D. $7^2 + x^2 = 2 \cdot 12$

C

The 12 is across from the right angle, so it is c and stands alone: $x^2 = 144 - 49 = 95$.

10. Ana finds the missing leg of a right triangle whose hypotenuse is 5 cm and whose other leg is 3 cm. She writes $3^2 + 5^2 = 34$, so the leg is about 5.8 cm. What did she do wrong? Find the leg.

She added the squares, and a leg cannot be longer than the 5-cm hypotenuse. Subtract: 25 − 9 = 16, so the leg is 4 cm.

the error named and the leg found

PART C. The Pythagorean Theorem in Real-World Problems

Lesson 4 · MA.8.GR.1.1 · 5 questions

11. A 10-ft ladder leans against a wall with its foot 6 ft from the wall. How high up the wall does it reach?

8 ft

the ladder is c: 100 − 36 = 64

12. A rectangular garden is 10 m long and 11 m wide. How long is a path straight across it from corner to corner, to the nearest tenth of a meter?

About 14.9 m (√221 m)

the diagonal is c: 100 + 121 = 221

13. A ramp rises 1 ft over a run of 9 ft along the ground. How long is the ramp, to the nearest tenth of a foot?

About 9.1 ft (√82 ft)

the ramp is c: 1 + 81 = 82

14. A park is a rectangle 9 m by 11 m. How much shorter is it to walk straight across the park from corner to corner than to walk along two of its sides? Give the answer to the nearest tenth.

About 5.8 m

along two sides 20 m; across √202 ≈ 14.2 m

15. A 9-ft ladder stands with its foot 2 ft from a wall. Will it reach a window ledge 8.5 ft up the wall?

Yes — it reaches √77 ≈ 8.8 ft up the wall, above 8.5 ft.

the height is a leg: 81 − 4 = 77

PART D. Distance on the Coordinate Plane

Lesson 5 · MA.8.GR.1.2 · 5 questions

16. Find the distance between the two points.

a. (−6, 2) and (5, 2)

11 units

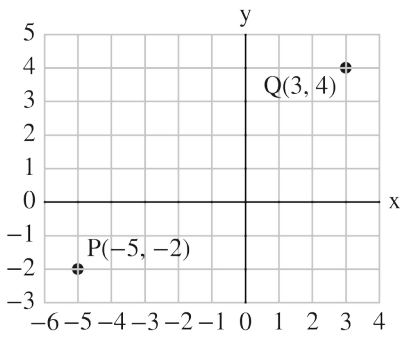
across zero: 6 + 5

b. (3, −7) and (3, 2)

9 units

across zero: 7 + 2

17. Find the distance between points P and Q.



10 units

across 8, up 6: 64 + 36 = 100

18. Find the distance between (−6, −1) and (1, 5), to the nearest tenth.

About 9.2 units (√85)

across 7, up 6: 49 + 36 = 85

19. How far is the point (−1, 6) from the origin? Round your answer to the nearest tenth.

About 6.1 units (√37)

legs 1 and 6: 1 + 36 = 37

20. What is the distance between (−2, −6) and (−2, 3)?

- ○ A. 3 units
- ● **B. 9 units**
- ○ C. 10 units
- ○ D. −9 units

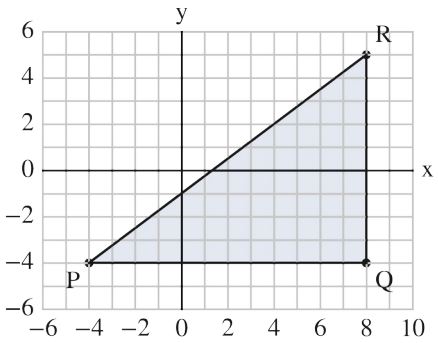
B

3 − (−6) = 9: 6 to the axis and 3 past it.

PART E. Coordinate-Plane Problems and Perimeter

Lesson 6 · MA.8.GR.1.2 · 4 questions

21. SSDD. Every part below is about one triangle on the grid: P(−4, −4), Q(8, −4), R(8, 5). Same surface, four different questions: a distance on a line (across zero), whether the corner at Q is square, the slanted side as a hypotenuse, and a similar copy (which needs QR, the shortest side, first) — the unit's moves on one triangle, so that a student learns to read the question and not the picture.



a. Find the length of PQ.

12 units

across zero: 4 + 8

b. Is triangle PQR a right triangle? How do you know?

Yes — PQ runs straight across and QR straight up, so they meet at a right angle at Q.

a horizontal side and a vertical side meet square (and $12^2 + 9^2 = 15^2$)

c. Find the length of PR.

15 units

the hypotenuse of legs PQ = 12 and QR = 5 − (−4) = 9: 144 + 81 = 225

d. A triangle similar to PQR has a shortest side of 3. How long is its longest side?

5

QR = 9 is the shortest side: k = 3 ÷ 9 = 1/3, and 15 × 1/3 — a copy smaller than the original

22. Find the perimeter of the triangle with corners (−4, 2), (6, 2) and (6, −1), to the nearest tenth.

About 23.4 units

10 + 3 + √109; √109 ≈ 10.440 — round once, at the end

23. On a town map in blocks, the school is at (−2, −4) and the park is at (2, 1).

a. How far is it from the school to the park along the streets?

9 blocks

4 across + 5 up

b. How far is it in a straight line? Round your answer to the nearest tenth.

About 6.4 blocks (√41)

16 + 25 = 41

24. A rectangle has corners (−3, −2), (5, −2), (5, 4) and (−3, 4).

a. Find its perimeter.

28 units

8 + 6 + 8 + 6

b. Find the length of one of its diagonals.

10 units

64 + 36 = 100

PART F. Similar Triangles and Indirect Measurement

Lessons 7–8 · MA.8.GR.2.4 · 6 questions

25. Are the two triangles similar? Show the three ratios.

a. Sides 5, 7, 8 and sides 15, 21, 24

Yes — 15/5 = 21/7 = 24/8 = 3.

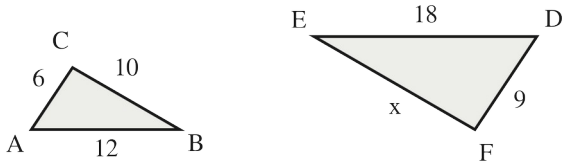
one k for every side

b. Sides 5, 8, 10 and sides 10, 16, 18

No — 10/5 = 16/8 = 2, but 18/10 = 1.8.

two equal ratios are not enough

26. Triangle ABC is similar to triangle DEF. Find x.



x = 15

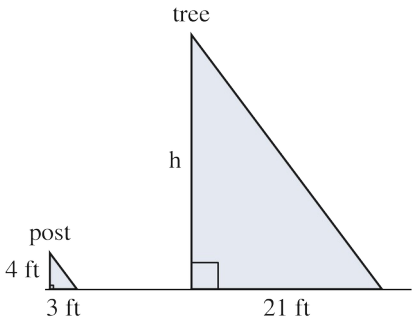
The names match AC with DF (6 with 9) and BC with EF, where x is: k = 1.5 and x = 10 × 1.5.

27. A triangle has sides 15, 20 and 30. A similar triangle's middle side is 8. Find its other two sides.

6 and 12

k = 8 ÷ 20 = 0.4 — the copy is smaller

28. A 4-ft fence post casts a 3-ft shadow. At the same moment, a tree casts a 21-ft shadow. How tall is the tree?



28 ft

k = 21 ÷ 3 = 7 from the shadows; 4 × 7

29. Kai's eyes are 5 ft above the ground. He stands 3 ft from a mirror on the ground and sees in it the top of a pole that stands 12 ft beyond the mirror. The two right triangles are similar. How tall is the pole?

20 ft

k = 12 ÷ 3 = 4; 5 × 4

30. Which triangle is similar to one with sides 6, 8 and 9?

- ○ A. 8, 10, 11
- ○ B. 12, 16, 20
- ● **C. 9, 12, 13.5**
- ○ D. 6, 8, 12

C

Every side × 1.5.

GRADE 7 ACCELERATED · UNIT 6 · PRACTICE TEST

Relationships in Triangles

18 points

Name_____
Date_____
Period_____

Show your work. One point for each lettered part.

Part 1. Triangle Inequality and the Converse

3 points

1. Which set of side lengths can form a triangle? *1 pt*

- ○ A. 4 cm, 7 cm, 11 cm
- ○ B. 12 cm, 5 cm, 6 cm
- ○ C. 6 cm, 8 cm, 13 cm
- ○ D. 3 cm, 10 cm, 15 cm

2. Select ALL of the sets of side lengths that make a right triangle. *1 pt*

- ☐ A. 17, 8, 15
- ☐ B. 5, 6, 7
- ☐ C. 20, 21, 29
- ☐ D. 7, 24, 26
- ☐ E. 4, 6, 8

3. Two sides of a triangle are 6 inches and 15 inches long. Which inequality shows every possible length x, in inches, of the third side? *1 pt*

- ○ A. $6 < x < 15$
- ○ B. $9 < x < 21$
- ○ C. $9 \leq x \leq 21$
- ○ D. $0 < x < 21$

Part 2. The Pythagorean Theorem *6 points*

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

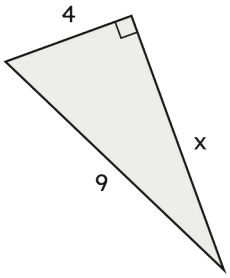
4. Find the missing side of each right triangle.

a. Its legs are 9 inches and 12 inches. How long is the hypotenuse? *1 pt*

b. Its hypotenuse is 61 meters and one leg is 60 meters. How long is the other leg? *1 pt*

5. A right triangle has legs of 7 centimeters and 13 centimeters. How long is the hypotenuse, in centimeters? Round your answer to the nearest tenth. *1 pt*

6. Which equation can be used to find x? *1 pt*



- ○ A. $4 + x = 9$
- ○ B. $4^2 + 9^2 = x^2$
- ○ C. $4^2 + x^2 = 2 \cdot 9$
- ○ D. $4^2 + x^2 = 9^2$

7. A rectangular garden measures 9 meters by 10 meters. A path runs straight across it from one corner to the opposite corner.

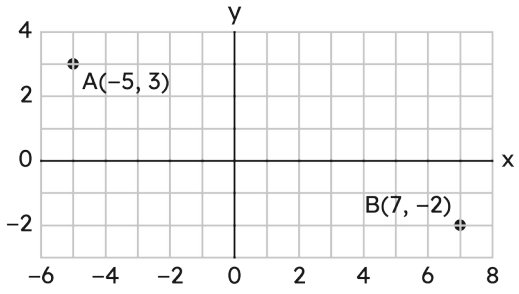
a. How long is the path, in meters? Round your answer to the nearest tenth. *1 pt*

b. How much shorter is the path than walking along two sides of the garden? Round your answer to the nearest tenth. *1 pt*

Part 3. Distance on the Coordinate Plane *6 points*

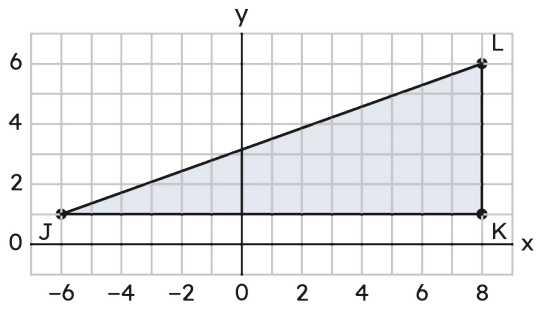
If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

8. Find the distance between points A and B. *1 pt*



9. Find the distance between (−1, 3) and (4, 1). Round your answer to the nearest tenth. *1 pt*

10. Triangle JKL has corners J(−6, 1), K(8, 1) and L(8, 6).



a. Find the length of side JK. *1 pt*

b. Find the perimeter of triangle JKL. Round your answer to the nearest tenth. *1 pt*

11. On a town map, each unit is one block. The library is at (−1, −5) and the pool is at (8, 7).

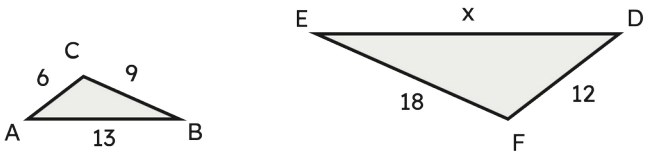
a. How far is it from the library to the pool along the streets? *1 pt*

b. How far is it in a straight line? *1 pt*

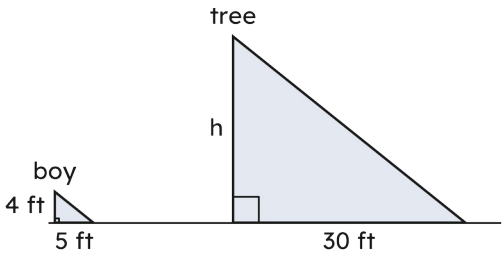
Part 4. Similar Triangles

3 points

12. Triangle ABC is similar to triangle DEF. Find x. *1 pt*



13. A boy 4 ft tall casts a shadow 5 ft long. At the same moment, a tree casts a shadow 30 ft long. How tall is the tree? *1 pt*



14. Which triangle is similar to a triangle with sides 6, 7 and 10? *1 pt*

- ○ A. 12, 14, 20
- ○ B. 10, 11, 14
- ○ C. 12, 14, 18
- ○ D. 6, 7, 12

GRADE 7 ACCELERATED · UNIT 6 · PRACTICE TEST

Relationships in Triangles

18 points **ANSWER KEY**

Name
Date
Period

Show your work. One point for each lettered part.

Part 1. Triangle Inequality and the Converse

3 points MA.8.GR.1.3

1. Which set of side lengths can form a triangle? *1 pt*

- ○ A. 4 cm, 7 cm, 11 cm
- ○ B. 12 cm, 5 cm, 6 cm
- ● **C. 6 cm, 8 cm, 13 cm**
- ○ D. 3 cm, 10 cm, 15 cm

C

The two shorter sides against the longest: 6 + 8 = 14, more than 13. A lies flat; B and D pass only when the wrong pair is added.

Scoring: 1 point.

2. Select ALL of the sets of side lengths that make a right triangle. *1 pt*

- ☒ **A. 17, 8, 15**
- ☐ B. 5, 6, 7
- ☒ **C. 20, 21, 29**
- ☐ D. 7, 24, 26
- ☐ E. 4, 6, 8

A and C

Square all three and test the longest as c: 64 + 225 = 289 = 17^2 (the longest listed first) and 400 + 441 = 841 = 29^2. Both and nothing else for the point.

Scoring: 1 point.

3. Two sides of a triangle are 6 inches and 15 inches long. Which inequality shows every possible length x, in inches, of the third side? *1 pt*

- ○ A. $6 < x < 15$
- ● **B.** $9 < x < 21$
- ○ C. $9 \le x \le 21$
- ○ D. $0 < x < 21$

B

15 − 6 = 9 and 15 + 6 = 21, never equal to either.

Scoring: 1 point.

Part 2. The Pythagorean Theorem

6 points MA.8.GR.1.1

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

4. Find the missing side of each right triangle.

a. Its legs are 9 inches and 12 inches. How long is the hypotenuse? *1 pt*

15 in

$c^2 = 81 + 144 = 225$

b. Its hypotenuse is 61 meters and one leg is 60 meters. How long is the other leg? *1 pt*

11 m

a leg: subtract — 3,721 − 3,600 = 121. Adding the squares gives a 'leg' longer than the hypotenuse.

Scoring: a 1 · b 1 = 2 points.

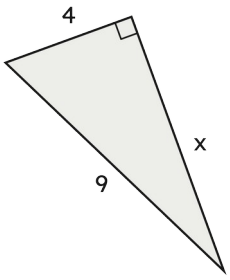
5. A right triangle has legs of 7 centimeters and 13 centimeters. How long is the hypotenuse, in centimeters? Round your answer to the nearest tenth. *1 pt*

14.8 cm (exactly √218 cm)

$c^2 = 49 + 169 = 218$; √218 = 14.76…

Scoring: 1 point.

6. Which equation can be used to find x? *1 pt*



- ○ A. $4 + x = 9$
- ○ B. $4^2 + 9^2 = x^2$
- ○ C. $4^2 + x^2 = 2 \cdot 9$
- ● **D.** $4^2 + x^2 = 9^2$

D

The 9 is across from the right angle, so it is c and stands alone: $x^2 = 81 − 16 = 65$.

Scoring: 1 point.

7. A rectangular garden measures 9 meters by 10 meters. A path runs straight across it from one corner to the opposite corner.

a. How long is the path, in meters? Round your answer to the nearest tenth. *1 pt*

13.5 m (exactly √181 m)

the path is the hypotenuse: 81 + 100 = 181

b. How much shorter is the path than walking along two sides of the garden? Round your answer to the nearest tenth. *1 pt*

5.5 m

9 + 10 = 19 m along two sides; 19 − 13.5. Follow-through on part (a).

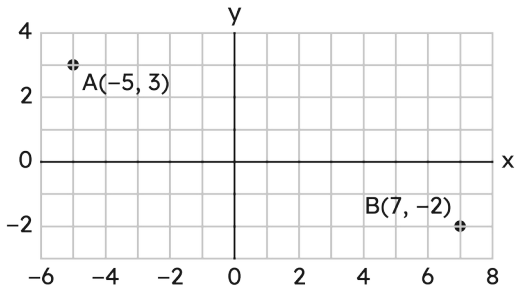
Scoring: a 1 · b 1 = 2 points.

Part 3. Distance on the Coordinate Plane

6 points MA.8.GR.1.2

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

8. Find the distance between points A and B. *1 pt*



13 units

Across 7 − (−5) = 12, down 3 − (−2) = 5: c^2 = 144 + 25 = 169.

Scoring: 1 point.

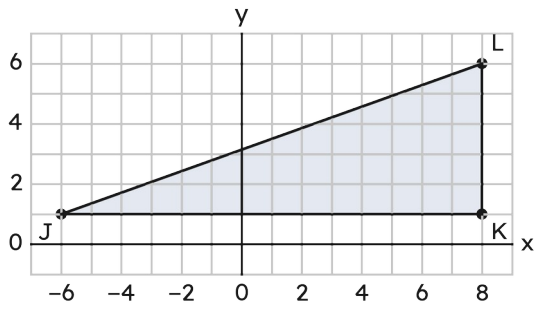
9. Find the distance between (−1, 3) and (4, 1). Round your answer to the nearest tenth. *1 pt*

5.4 units (exactly √29)

Across 5, down 2: c^2 = 25 + 4 = 29.

Scoring: 1 point.

10. Triangle JKL has corners J(−6, 1), K(8, 1) and L(8, 6).



a. Find the length of side JK. *1 pt*

14 units

Across zero: 6 + 8 — never 8 − 6.

b. Find the perimeter of triangle JKL. Round your answer to the nearest tenth. *1 pt*

33.9 units

KL = 5; JL = √(196 + 25) = √221 ≈ 14.866; 14 + 5 + 14.866 = 33.866 — round once, at the end. Follow-through on part (a).

Scoring: a 1 · b 1 = 2 points.

11. On a town map, each unit is one block. The library is at (−1, −5) and the pool is at (8, 7).

a. How far is it from the library to the pool along the streets? *1 pt*

21 blocks

The streets follow the grid: 9 across and 12 up.

b. How far is it in a straight line? *1 pt*

15 blocks

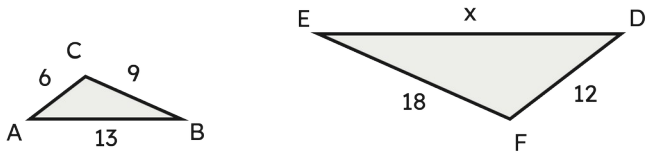
The hypotenuse of legs 9 and 12: 81 + 144 = 225.

Scoring: a 1 · b 1 = 2 points.

Part 4. Similar Triangles

3 points MA.8.GR.2.4

12. Triangle ABC is similar to triangle DEF. Find x. *1 pt*

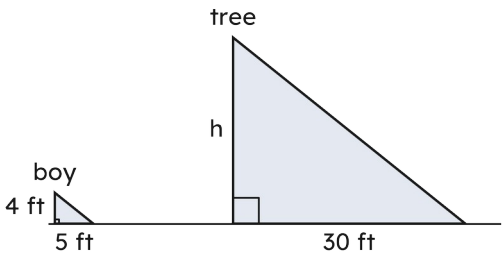


x = 26

The names match AC with DF and BC with EF — 12 with 6 and 18 with 9, as by size — so k = 2, and x is on DE, which matches AB: 13 × 2. 39 matched by place on the page (18 with the 6); 13 took similar as equal; 19 added the difference 12 − 6.

Scoring: 1 point.

13. A boy 4 ft tall casts a shadow 5 ft long. At the same moment, a tree casts a shadow 30 ft long. How tall is the tree? *1 pt*



24 ft

k = 30 ÷ 5 = 6 from the shadows; 4 × 6. 37.5 matched height and shadow crosswise; 120 multiplied by the boy's height, not by k.

Scoring: 1 point.

14. Which triangle is similar to a triangle with sides 6, 7 and 10? *1 pt*

- ● **A. 12, 14, 20**
- ○ B. 10, 11, 14
- ○ C. 12, 14, 18
- ○ D. 6, 7, 12

A

Every side × 2: 12/6 = 14/7 = 20/10.

Scoring: 1 point.

Score Tracker

Benchmark	Questions	Possible	Earned
MA.8.GR.1.3	1, 2, 3	3	

Benchmark	Questions	Possible	Earned
MA.8.GR.1.1	4, 5, 6, 7	6	
MA.8.GR.1.2	8, 9, 10, 11	6	
MA.8.GR.2.4	12, 13, 14	3	
Total		**18**	

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page. Nothing is weighed and nothing is arguable: apply the part's operation to the student's own earlier value and tick if that is what is written.